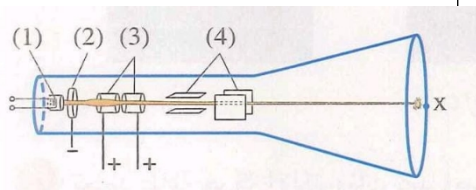


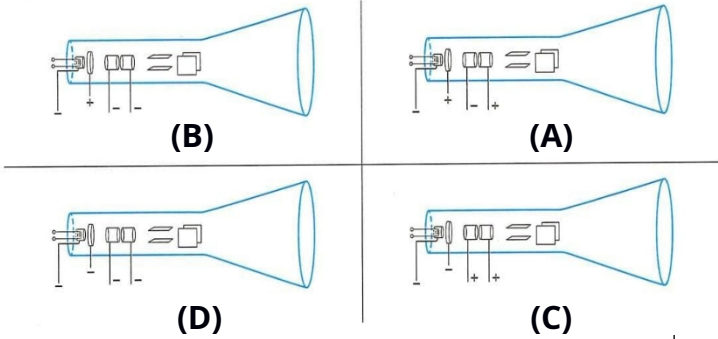
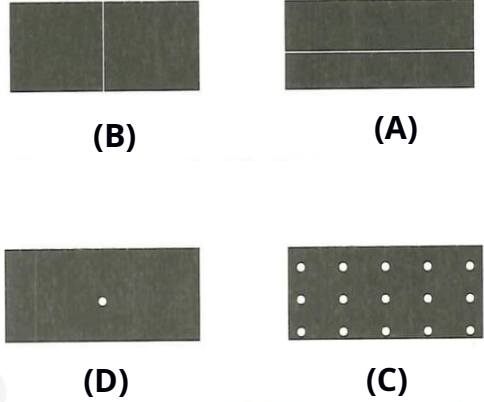
المعرفة	الكود
Thermionic emission	TPK01

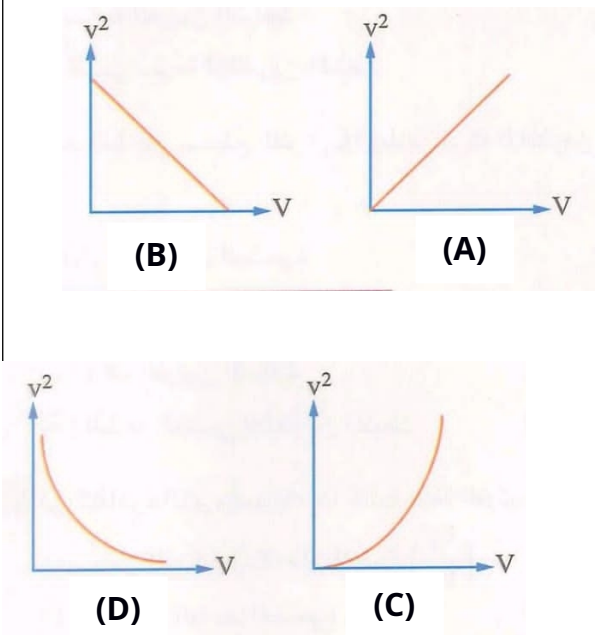
1- Choose the correct answer:

#	Question	Choices	Answer
1	The cathode rays consist of	a. Electrons b. Photons c. Electromagnetic Wave d. Gamma	a
2	Cathode rays have the energy of	a. $\frac{1}{2} m_e V^2$ b. $m_e V$ c. $2 m_e V$	a
3	The process in which electrons are emitted from a metallic surface when heat given to it is called	a. Photoelectric effect b. Thermionic effect c. Compton effect d. Black body radiation	b
4	CRT stands for	a. Common Ray Tube b. Common Running Tube c. Cathode Ran Tube d. Cathode Ray Tube	d
5	In the C.R.T when the grid voltage changes from -20 V to -50 V.....	a. Intensity of illumination increases on fluorescent screen. b. Intensity of illumination decreases on fluorescent screen. c. The deflection of electron beam increases. d. The deflection of electron	b

		beam decreases.	
6	From the properties of beam of C.R.T.....	a. Electromagnetic waves. b. With constant speed. c. Charged particles. d. Uncharged particles.	b
7	The opposite figure represents C.R.T, which of the following parts represents..... A. The source of electrons..... B. The part that responsible for changing the position of collision of electron beam with fluorescent screen..... C. The illumination intensity is controlled by At point X.	(A): a. Part (1). b. Part (2). c. Part (3). d. Part (4). (B): a. Part (1). b. Part (2). c. Part (3). d. Part (4). (C): a. Part (1). b. Part (2). c. Part (3). d. Part (4).	(A): a (B): d (C): b



8	Which of the following figures represents the correct voltage that is used in C.R.T.....		
9	In the C.R.T when the filament is damaged	a. Intensity of illumination increases on fluorescent screen. b. Intensity of illumination decreases on fluorescent screen. c. The fluorescent screen doesn't illuminate. d. The deflection of electron beam decreases.	c
10	In the C.R.T when the grid is not connected to any electric signal	a. It is hard to control the path of electric beam into the screen. b. The fluorescent screen doesn't illuminate. c. The electric beam reflects to cathode. d. The illumination remains constant.	d
11	Which of the following figures represents the final image that formed in the screen of C.R.T when the deflection system doesn't work.....		d
1	The kinetic energy for the	a. The surface area of cathode.	c

2	electrons is depend on when it reaches the Anode in C.R.T.	b. Work done for Anode. c. Potential difference between two plates of deflection system. d. Potential difference between cathode and anode.	
1 3	In C.R.T the velocity of electron is v , when the electron is accelerated by potential difference with V , if the potential difference is increased to $2V$, son the velocity will become.....	a. $2v$ b. $\sqrt{4v}$ c. $4v$ d. $1\sqrt{2}v$	c
1 4	Which of the following graphs represents the relationship between v^2 for the electron and V in C.R.T.....	 (B) (A) (D) (C)	a
1 5	If the potential difference between anode and cathode in C.R.T is 1000 V : so,... A. K.E for the electron is B. The maximum velocity of	(A): a. $1.6 \times 10^{-15} J$ b. $1.6 \times 10^{-16} J$ c. $1.6 \times 10^{-17} J$ d. $32 \times 10^{-18} J$ (B) : a. $1.88 \times 10^6 m$ b. $1.88 \times 10^7 m$	(A) b (B) b

	liberated electron from cathode when it reaches anode equals.....	c. $3.52 \times 10^5 m$ d. $3.52 \times 10^4 m$	
1 6	The source of thermal energy in the C.R.T is.....	a. Anode. b. Cathode. c. Filament. d. Fluorescent screen.	c
1 7	The reason for the kinetic energy of the electron is.....	a. Velocity. b. Density c. Mass of the electron. d. Potential difference between cathode and anode.	d
1 8	It is used to accelerate the electron beam towards the screen.....	a. Cathode. b. Anode. c. Grid. d. Filament.	b
1 9	To increase to velocity of electron in C.R.T for four times.....	a. Increase K.E for four times. b. Increase the mass of electron to double. c. Increase the potential difference to double. d. Increase the potential difference for four times.	c
2 0	To decrease the intensity of electron beam	a. Increase the voltage of grid. b. Decease the voltage of anode. c. Increase the voltage of anode. d. Decrease the voltage of grid.	a
2 1	Which part of C.R.T is responsible for the deflection of electron beam.....	a. Vertical plates. b. Horizontal plates. c. Anode. d. A,B.	d

2- Mention the application of

No	Question	Answer
1	The electron gun in CRT	It is an instrument to produce a beam of electrons at high speed
2	CRT	used in TV and computer monitors
3	Deflection system in CRT	Deflect the path of the electron
4	fluorescent screen in CRT	Emits green light when the electron strikes it
5	Filament in CRT	It is used to heat up the cathode
6	Cathode in CRT	Emission of electrons
7	Grid in CRT	Control the number of electrons in the electron beam
8	Anode in CRT	It is used to accelerate and focus the electron beam towards the screen

3- Mention the factor affecting on

No.	Question	Answer
1	KE of the freed electron in CRT	Accelerating voltage or velocity of the electron
2	the velocity in freed electron in CRT	Accelerating voltage or kinetic energy

4- Solve the following:

No	Question	Answer
.		

1	An electron moving at a constant speed 4.4×10^5 m/sec, what is its kinetic energy?	$KE = \frac{1}{2} m_e v^2 = \frac{1}{2} m_e (4.4 \times 10^5)^2 = 8.82 \times 10^{-20} J$
2	Calculate the speed of an electron that has kinetic energy of 120 eV.	$(K.E. = eV = \frac{1}{2} m_e v^2)$ $120 \times e = \frac{1}{2} m_e v^2 \Rightarrow v = 6497051 \text{ m/s}$
3	calculate the potential difference which accelerated an electron from rest with speed equals 9.796×10^6 m/s	$(K.E. = eV = \frac{1}{2} m_e v^2)$ $V = \frac{\frac{1}{2} m_e v^2}{e}$ $V = 272.8 \text{ volts}$

5- complete the following sentences:

No	Question	Answer
1	There are many ways to eject (escape) electrons out of a materialand	Thermionic emission – photoelectric effect
2	Emission of electrons from a heated metal's surface (by heating) is called	Thermionic emission
3	 is application on the thermionic emission.	Cathode ray tube (CRT)
4	C.R.T is used to make and Monitors.	TV – computer
5	 is the attractive force between the positive ions and free electrons in metal which make	Surface potential barrier

	the electrons cannot leave it.	
6	Operation stages for C.R.T areand	Electron gun – deflection system
7	 is used to control the number of electrons of the electron beam .	grid
8	 is used to accelerate the electron beam towards the screen.	Anode
9	The stage at which the beam at high speed.....	Electron gun
10	emits green light when the electron strike it.	Fluorescent screen

6- Mention the scientific term

No .	Question	Answer
1	emission of electrons from a heated metal's surface (by heating)	Thermionic emission
2	emission of electrons from a metallic surface when the light energy falling on it	photoelectric effect
3	It is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it	Surface potential barrier